

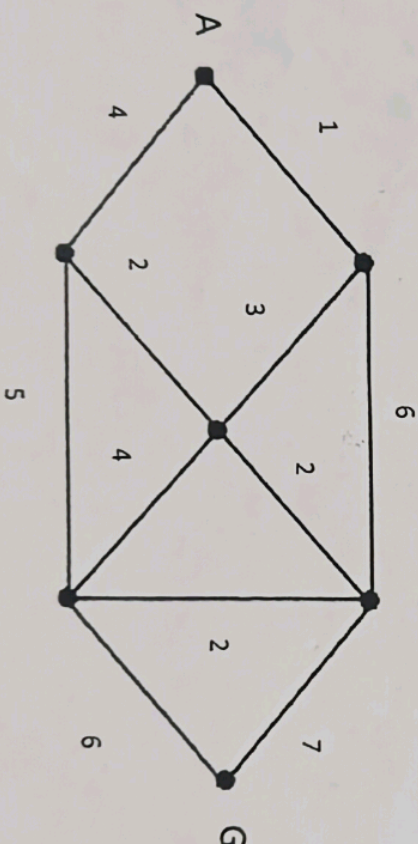
DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Program: B.Tech. Semester-IV
Course Title: Discrete Mathematics and Statistical Methods
Course Code: MAIC 201

Paper: REGULAR and REAPPEAR
Time: 06:00PM – 06:50 PM
M.M. 15

Note: Attempt all the Questions. Each question carries equal marks.

6. Consider the set R of all real numbers under the relation $\leq$ (usual "less than or equal to"). Is $(R \times R, \leq)$ is a poset under this relation?
7. Show that an undirected graph possesses a Eulerian path if and only if it is connected and has either zero or two vertices of odd degree.
8. Find the shortest path between A and G in the graph below.



9. Construct a truth table for $(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$. Find out whether it is tautology or not.
10. The heights of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% significance level, assuming that for 9 degrees of freedom $P(t > 1.83) = 0.05$. [$t_{0.10} = 1.833$ for 9 d.f.].